

# Tomas Benedicto

Head of Payload Architects · Airbus Defence and Space · Toulouse, France
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## I architect space systems end to end.

Aerospace engineer with 10 years in industry. I develop systems that span the space, ground and user segments, from flight hardware to concepts of operations. Today I lead Airbus's **payload architects and mission managers**: we support bids and design the next generation of telecom payloads, across all orbits.

| 10 | 10+ | 8+ | 4 |
|---|---|---|---|
| years in the space industry | space programmes and products | global operator customers | working languages |

PROGRAMMES I'VE WORKED ON

OneSat flexible GEO telecom satellite (more than 8 global operators) ✦ OneWeb telecom constellation ✦ Galileo Second Generation ✦ Eurostar NEO high-powered GEO telecom satellite ✦ AstroBus-S Earth observation ✦

## 01 What I bring

Three things set me apart: I've worked at every level of engineering, on every major satellite subsystem, and I now lead a team that delivers.

### Every level

Analyst, designer, tester, architect, chief engineer, project manager, and now Head of Payload Architects. I know what each role needs because I've done it, and I take ownership from the first calculation to the final decision.

### Every subsystem

Mechanical structures, electric propulsion, RF and digital telecom payloads. I see how a decision in one subsystem lands in the others, so I can design the system end to end.

### Leadership & delivery

From Center of Competence board member to head of a team of 10+. I set the roadmap, cost and plan the work, manage subcontracting and empower people to deliver on time, on cost and on quality. I build trusted customer relations with 8+ global operators and key suppliers.

## 02 Experience

### Head of Payload Architects

Current
Aug 2026 – Present
Toulouse, FR

**Airbus Defence and Space · Telecom satellite payloads · promoted Aug 2026**

- Leading the payload architects and mission managers shaping the future of Airbus telecommunications satellite payloads: building technical excellence, developing people, and delivering competitive solutions that create lasting value for commercial and institutional customers.
- Supporting bids and designing next-generation payloads across all orbits; costing and planning the team's work and managing its subcontracting, with 8+ global operator customers and key suppliers.

### Telecom Payload Systems Lead

Dec 2023 – Jul 2026
Toulouse, FR

**Airbus Defence and Space · OneSat**

- Technical lead defining the concept of operations of a flexible digital telecommunications system spanning the space, ground and user segments, leading a team of 10+ payload architects.
- Project manager for the payload, running its industrial contracts, schedule and risk.
- Chief engineer and technical authority of the OneSat Payload Configuration Manager, spanning space segment, ground segment and in-orbit operations.
- Customer relations and industrial contracts with 8+ global operators and several key suppliers, under tight cost, schedule and technical constraints.

### Satellite Propulsion Lead Engineer & Project Manager

Jun 2021 – Jun 2024
Toulouse, FR

**Airbus Defence and Space · Galileo Second Generation and a new European constellation**

- Propulsion lead engineer for ESA's next-generation Galileo navigation satellites (Jun 2021 – May 2024): architecture of the electric propulsion subsystem and its mechanisms, verification and validation, and in-flight operations planning, reporting to the satellite chief engineer.
- Project manager (Jun 2021 – Jun 2024): led the end-to-end development of a new plasma propulsion assembly for an innovative European satellite constellation.
- Designed a propulsion mechanism from scratch to satellite assembly in three years.
- Worked across propulsion, electrical, thermal and avionics engineering; flowed system requirements down to equipment and controlled compliance, deviations and waivers.
- Managed suppliers and chaired critical design reviews and non-conformance boards with international stakeholders; delivered on time, on cost and on quality.

### Spacecraft Platform Design & Analysis Engineer
**Airbus Defence and Space**
Oct 2017 – Jun 2021
Toulouse, FR · Stevenage, UK

- Designed, justified and tested satellite platforms and thermo-mechanical products for telecom satellites in LEO and GEO and for Earth observation satellites, OneWeb and OneSat among them.
- Optimised the design of OneSat's novel structure and of the interfaces between satellites on the launcher and with the launcher itself.
- Co-authored Airbus's international, ECSS-aligned industry standard for modelling and optimising space structures.

### Aerospace Engineer, Structural Optimisation
**Airbus Defence and Space**
Apr 2017 – Sep 2017
Toulouse, FR

- Structural optimisation of spacecraft structures, working to ECSS standards.

### Trainee, Structures & Mechanisms Division
**European Space Agency · ESTEC**
Jul 2015 – Sep 2015
Noordwijk, NL

- In the Spacecraft Dynamics Analysis team, verified the Ariane 5 to satellite interface in preparation for the first Ariane 5 launch of Galileo satellites.

## 03 Leadership & R&D

2022 – 2024 · Airbus DS
**Focal point, nuclear propulsion for space**
Airbus's focal point for system-level R&D with nuclear-industry partners toward an ESA demonstrator targeted for 2035.

2018 – 2021 · Airbus DS
**Center of Competence board member**
Co-authored industry standards for finite element modelling and structural optimisation.

2018 · ESA & Airbus DS
**ECSS MET conference paper**
Co-author of a scientific paper on design processes for structural parts made by additive layer manufacturing.

## 04 Education & certifications

### M.Sc. Aerospace Mechanics & Avionics
**ISAE-SUPAERO · Toulouse**
2015 – 2017
Exceptional Academic Achievement Award 2017
Specialised in aerospace structures. Research on the failure modes of high-strength fibre composites using infrared techniques. Master students' representative on the university board and student representative of the Royal Aeronautical Society's Toulouse branch.

### B.Sc. Aerospace Engineering
**TU Delft · Delft**
2012 – 2015
Awarded with Distinction
A broad foundation in aerospace engineering: aerodynamics, flight mechanics, structures and materials, propulsion and spaceflight, closed by a multidisciplinary team design project.

### Systems Engineering & Excellence
**Airbus DS · Toulouse, Ottobrunn & Getafe**
2026
Airbus's chief-engineer track: leadership, system accountability and technical decision-making.

### Project Management Professional (PMP®)
**PMI**
Oct 2024

### Architecture & Systems Engineering: Models & Methods to Manage Complex Systems
**MIT xPRO · professional certificate**
Jan – Jul 2021
The systems engineering framework and MBSE for developing complex systems and managing uncertainty.

## 05 Toolkit

TECHNICAL
End-to-end systems engineering · Telecom payload · RF · Digital signal processing · Electric propulsion · Electrical · Thermal · Avionics · Mechanics & materials · LEO constellations · AIT · V&V · MBSE · ECSS · Python

LEADERSHIP
Project management · Contract leadership · Customer relations · Negotiation · Facilitation · Strategic thinking · Clear communication

LANGUAGES
Spanish — Native
French — Native
English — Bilingual
Dutch — Professional
German — Elementary

OFF DUTY
Football · Windsurfing · Piano